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CSCI 4370

Preliminary Report

Title: UGA Parking Spot Tracker

Problem: Parking on UGA's campus can be difficult to plan because students often do not know how full a parking lot is until they arrive. This can waste time, add stress, and cause students to drive between multiple lots looking for an available spot. Information about parking availability is also not always updated in real time, making it harder for students to make quick decisions before heading to campus.

Solution: This application addresses the parking availability problem by creating a crowd-sourced, database-backed system where students can view UGA parking lots and report current lot conditions. Users can see how packed a lot is, whether open spots are visible, and when the latest report was submitted. By collecting and displaying this information in one place, the application helps students make more informed parking decisions before driving to a lot.

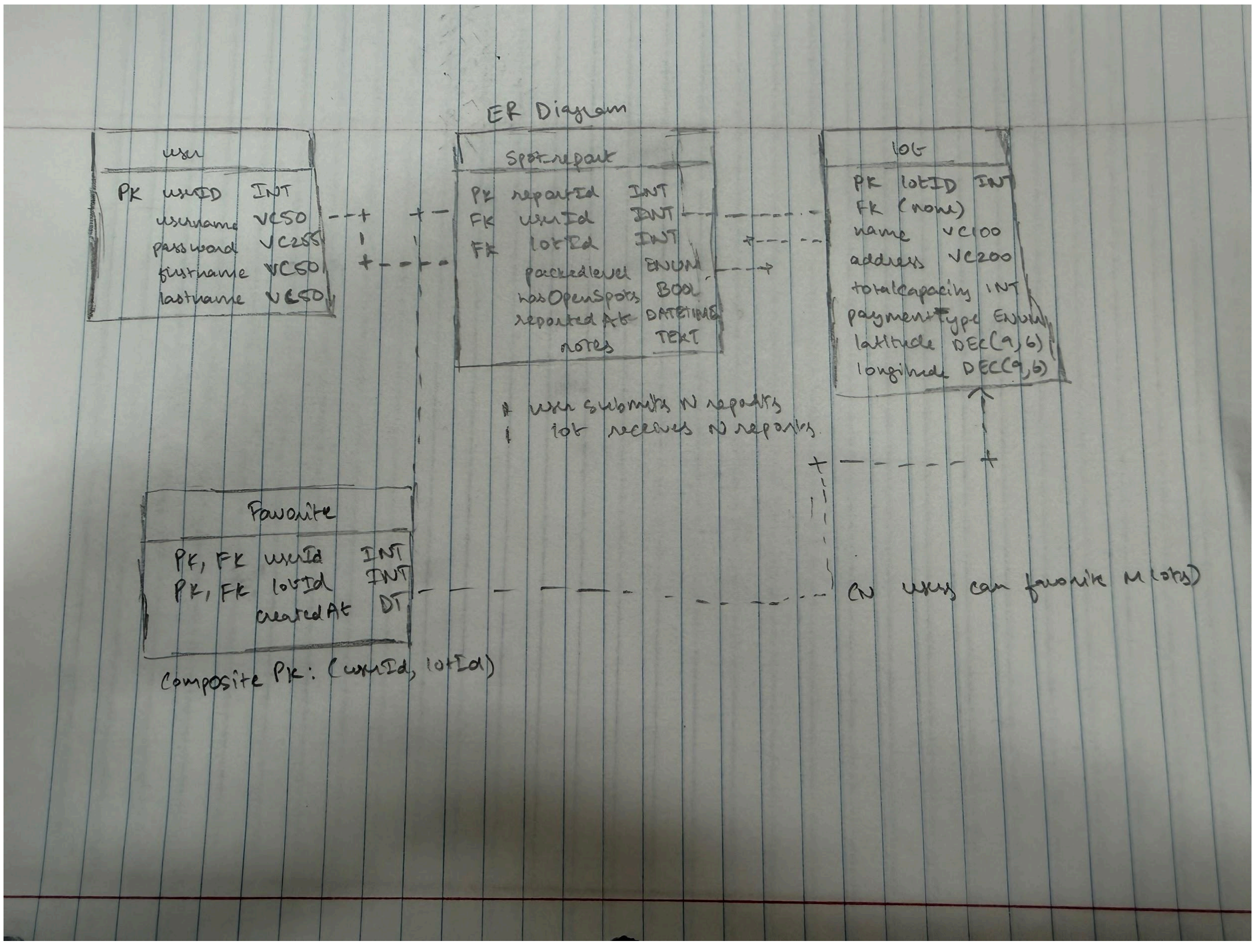
UGA Parking Spot Tracker is a web app that includes the following functions:

User sign up and authentication
Viewing UGA-affiliated parking lots
Searching for parking lots by name or address
Filtering lots by payment type
Viewing detailed parking lot information
Submitting crowd-sourced parking availability reports
Viewing report history for a selected lot
Adding, editing, and deleting parking lot records

Updating user profile information

Viewing analytics such as busiest hours, most available lots, and top reporters

ER DIAGRAM:



User Interface: These are key aspects of our web app that we aimed to include:

Signup Page - creates a new user account

Login Page - authenticates a user

Lots Page - displays UGA parking lots and supports search and filtering

Lot Detail Page - displays information about a selected lot, including location, capacity, payment type, and latest availability

Add/Edit Lot Page - allows parking lot information to be created or updated

Report Page - allows users to submit a parking availability report for a lot

Lot Reports Page - displays recent crowd-sourced reports for a selected parking lot

Profile Page - allows users to view and update their account information
Analytics Page - displays useful parking trends such as busy hours, available lots, and top reporters

Technology: We chose tools that support a database-backed web application and matched the technologies used throughout the course/project. Java and Spring Boot handle the backend logic, routing, authentication, and database access. Mustache templates, HTML, CSS, and JavaScript are used for the frontend interface. MySQL stores users, parking lots, reports, and related data.

Frontend: HTML, CSS, JavaScript, Mustache
Backend: Java, Spring Boot
Database: MySQL
Environment: Maven, Visual Studio Code
Version control: Git